Self-Auditing Formal-Systems MCP Kernels for Autonomous Agents:

Empirical Soundness Defects, Hash-Collision Pruning, and Quotient-Homomorphism Proofs

on the Sokoban Eval Frontier

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Abstract

When large language model (LLM) agents author both a combinatorial solver and its accompanying verification script, passing unit tests frequently masks silent soundness failures. During a multi-agent evaluation on the Sokoban Eval Frontier Challenge across Jonah Ostroff's 66-room *Heroes of Sokoban* trilogy, an initial agent-authored Model Context Protocol (MCP) server (`soko-verified-mcp v0.1.0`) passed all six baseline tests and reproduced a live record 84-key trajectory on Room `III-20` (surpassing the 85-key holdout target and 145-key model baseline). Yet cross-agent adversarial review uncovered six soundness and generality defects (`F-01`–`F-06`), including two critical verification failures: (1) an optimistic boolean initialization that silently emitted `VERIFIED` certificates from a single interpreter whenever C++20 compilation was absent (`F-01`), and (2) lossy tag-only transposition-table pruning in lower-bound search that suffered **22,658 live 64-bit hash collisions** within 35 seconds across 2.766×10^7 exact states on Room `III-14` (`F-04`).

To eliminate these failure modes structurally, `soko-verified-mcp` (`v0.2.0`–`v0.3.0`) introduces a domain-general, 16-tool Formal-Systems MCP Kernel (`formal_*` and `soko_*`). The kernel enforces AST-sandboxed pure execution over exact integers and rationals, blocks alpha-renamed copy-paste tautologies via structural AST skeleton hashing, mandates negative-control rejection and planted-mutation kill gates before unlocking search, and restricts lower-bound certificates to machine-checked Quotient Graph Homomorphisms, Inductive Potentials, and Houdini greatest-fixpoint invariants. Across three evaluation domains, the hardened server certifies the 84-key chiral-mirror record on `III-20`, raises the collision-free exact-state lower bound on `III-14` from 9 to 35 across 1.108×10^8 verified transitions via a 3.97×10^7-state 2-Atom Quotient Pattern Database, extracts a 5-step split-brain counterexample in an unfenced dual-gateway lease protocol before proving global safety across all 486 syntactic states under monotonic epoch fencing, and closes a precedence-constrained setup-cost scheduling benchmark at LB = UB = 19. Following Foundation-0 epistemic rules, this paper separates the genuine systems contribution—mechanical anti-tautology gates for agentic verification—from the classical nature of the small-state protocol and scheduling benchmarks.

Epistemic Summary: Claims in this preprint follow the four-tier Foundation-0 taxonomy: `PROVEN` (exhaustive deductive or closed-universe finite mathematical proof), `VERIFIED` (finite deterministic witness replayed across ≥ 2 zero-shared-code checkers with SHA-256 checker pins), `KNOWN` (external published baseline), and `CONJECTURE` (unproved search hypothesis).

*Prepared under the Foundation-0 Charter (Principles 0.2–0.6: explicit human sign-off, AI role disclosure, calibrated epistemic claim labels, and open correction of defects) and the Proven-Commons Verified Bar. Nothing in this preprint has been externally peer-reviewed.

1 Verification Illusions in Agent-Authored Solvers

Combinatorial planning benchmarks expose a recurring asymmetry between discovering candidate trajectories and certifying mathematical claims. On the Sokoban Eval Frontier Challenge (*Eight moves to the exit*), frontier language models evaluated across the 66 rooms of Jonah Ostroff's PuzzleScript trilogy—*Heroes of Sokoban Part I*, *Part II: Monsters*, and *Part III: The Bard and The Druid*—achieved proven optimality in 44 of 66 rooms with a best composite segment total of 3,819 inputs against a 3,000-key trilogy lower-bound floor and 3,064 keys across the 22 holdout routes [1, 2].

Unlike classical single-agent Imabayashi Sokoban, the trilogy composes six coupled hero movement laws under PuzzleScript's synchronous rule-rewriting engine [3]:

1. **Fighter (`F/1`):** Pushes colinear chains of arbitrarily many `Moveable` entities (crates `*`, dormant heroes `1..6`, and monsters) whenever an empty cell terminates the ray.
2. **Thief (`T/2`):** Steps into an empty cell and pulls one orthogonally trailing `Moveable`.
3. **Wizard (`W/3`):** Fires a projective warpball that swaps positions with the first `Moveable` struck, or advances one empty cell if striking a `Static` barrier (`#`, closed doors, monster gates `M`, or Part III board edges).
4. **Priest (`P/4`):** Protects itself and orthogonally adjacent heroes—both when active (`P`) and when dormant (`4`)—against simultaneous `Late`-phase Goblin/Dragon `Dying` rules and Druid petrification.
5. **Bard (`B/5`, Part III):** Projects a 5×5 Chebyshev `Note` aura ($\max(|\Delta x|, |\Delta y|) \leq 2$) computed at the *conclusion of the preceding turn*; on movement, Bard and all charmed `Moveable` objects translate rigidly to a simultaneous fixed point, reverting via `Cancel` if Bard itself cannot advance.
6. **Druid (`D/6`, Part III):** Transmutes adjacent crates or goblins into walls ($\{\texttt{*}, \texttt{O}\} \to \texttt{\#}$), walls (including outer boundary walls) into crates ($\texttt{\#} \to \texttt{*}$), and unshielded dormant heroes into walls, where transmuted walls count as `Weighing` on yellow and purple pressure switches.

When autonomous coding agents translate rich rule systems into fast C++ or Python search engines, subtle semantic drift is common: evaluating Bard's aura in Manhattan rather than Chebyshev norm, computing charm masks before rather than after a move, cycling `Action` (`A`) in `OBJECTS` order rather than `LEGEND` order, or resolving Goblin combat sequentially rather than simultaneously. More dangerously, when the same agent writes the test harness, passing tests can create an *illusion of formal verification*. This paper documents what survived inside a live record-setting MCP server (`v0.1.0`) and how a minimal, self-auditing MCP protocol (`v0.3.0`) mechanically eliminates those failure modes.

2 Empirical Soundness Audit: Six Defects Caught in Review

Following the pilot clear of Room `III-20` in 84 keys on branch `feat/soko-verified-mcp` (`v0.1.0`), three review lanes (`fdn0-eval` in PR #1, `aeva` in PR #2, and `gem-nexus` in `v0.3.0`) audited the codebase against the Foundation-0 Repository Standard [4] and the Proven-Commons Verified Bar [6]. Table 1 catalogs the defects uncovered despite a 100% passing `v0.1.0` test suite.

2.1 Anatomy of Two Silent Verification Hazards

Two defects in Table 1 illustrate why code review by independent verification seats is indispensable even when tests pass:

Table 1: Adversarial soundness and specification defects uncovered in `soko-verified-mcp` `v0.1.0`–`v0.2.0` and resolved in `v0.3.0`.

ID	Severity	Root Cause in Baseline Code	Mechanical Resolution in `v0.2.0`–`v0.3.0`
`F-01`	**Critical**	`verify_route_dual` initialized `differential_match = True` prior to `if cpp_res is not None:`, emitting `VERIFIED` on a single oracle if `g++` was missing or $W \cdot H > 512$.	Added zero-shared-code 1D bitmask `Flat BitmaskOracle`; initialized `differential_match = False`; enforced ≥ 2 independent engines.
`F-02`	**High**	Differential check compared `final_ascii` only at terminal step $t = L$, collapsing open doors (`YO`/`PO`) to `.` and ignoring hidden Bard `Note` bitmasks on $t < L$.	Folded 64-bit FNV-1a full-state per-frame digests (`traj_fnv64`) over $(\text{objs}, \text{notes}, y_{\text{open}}, p_{\text{open}})$ at every frame $t = 0 \ldots L$ across all three engines.
`F-03`	**High**	`SHA256SUMS` was unchecked at runtime; certificates omitted checker hashes (Issue #14); mutation audit tested only one engine.	Added path-confined `soko_verify_tcb_lock`, embedded SHA-256 `checker_pins` in all certificates, and checked cross-oracle parity across all mutation vectors.
`F-04`	**Critical**	`prove_lb` pruned IDA* branches on a hash-tag hit without comparing full `DynState`; admissible heuristic collapsed to $h_0 = 9$ on `III-14` and $h_0 = 1$ with Wizard.	Stored full `DynState`/`Packed14` records in exact-state transposition/A* tables (preventing 22,658 live collisions on `III-14`); added `PairAtomQuotientPDB` and `WizardRayGoalBound`.
`F-05`	**Arch.**	All 7 MCP tools hardcoded Sokoban ASCII grids and `U/D/L/R/A` moves; lacked 15×7 trailing-column variant of `III-14`.	Built 7 universal AST-sandboxed `formal_*` tools (`formal_kernel.py`), `soko_capabilities`, and both 14×7 and 15×7 `III-14` benchmarks.
`F-06+`	**Med/Low**	`README.md` mistyped alternate `III-20` SHA-256 (`4c99...` vs. `4edff1ad8c90`); `I-1` listed 8 steps instead of 9 (`RRRRRRRRR`); `prove_floor` lacked fail-closed witness replay.	Fixed `III-20` digest and 9-step `I-1` baseline; made `prove_floor_verified` fail closed (`INCOMPLETE_CPP_OUTPUT`, `REJECTED_INVALID_WITNESS`).

1. Optimistic Boolean Defaults (`F-01`). In `v0.1.0`, `verify_route_dual` ran the Python reference oracle, attempted to run the C++20 binary, and set `differential_match = True` *outside* the conditional block checking whether C++20 succeeded. On any container lacking `g++`/`clang++`, or on any grid exceeding the fixed 512-cell C++ buffer, the server silently skipped differential replay while returning `"epistemic_status": "VERIFIED"`. All six unit tests passed because the authoring environment had `g++` installed and tested rooms with $W \cdot H \leq 195$. Initializing `differential_match = False` (fail-closed) and pairing `PuzzleScriptOracle` with a zero-shared-code 1D integer bitmask interpreter (`FlatBitmaskOracle`) guarantees genuine two-engine replay in pure Python and three-engine replay whenever C++20 is compiled.

2. Birthday-Paradox Hash Collisions in Exact Lower-Bound Provers (`F-04`). Heuristic game engines routinely prune transposition subtrees on 48-bit or 64-bit hash tags without storing full states, because rare false prunes scarcely change playing strength. In a *mathematical lower-bound prover*, however, a single false prune invalidates the proof: if state s_{opt} on an optimal path collides in hash tag with an earlier state s_{dead} recorded with greater remaining budget, the prover prunes s_{opt} and can emit an unsound lower bound $\text{OPT} > B$.

For N distinct visited states hashed into b-bit tags (or an open-addressed table of $M = 2^{18}$ slots), birthday-paradox collisions scale quadratically as $\mathbb{E}[\text{collisions}] \approx N^2/2^{b+1}$, while slot-index collisions occur continuously at high load factors. Instrumenting full `DynState` and 112-bit `Packed14` equality checks on Room `III-14` revealed:

- **11 live hash collisions prevented** in a 6-second IDA* run (802,816 nodes; 3,928,470 edge checks) in the 2^{18}-slot table;
- **1,014 live full-64-bit hash collisions prevented** in a 10-second multi-core bucketed A* run (4.14×10^6 transitions); and
- **22,658 live full-64-bit hash collisions prevented** in a 35-second exact-state A* run (2.766×10^7 unique `Packed14` states; 1.108×10^8 transitions).

Every one of those 22,658 events was a genuine 64-bit hash collision between two distinct reachable configurations of Room `III-14`. Full-state comparison reduces false hash-collision pruning to identically zero.

3 Zero-Trust Formal-Systems MCP Kernel Architecture

`soko-verified-mcp v0.3.0` exposes 16 MCP tools over stdio JSON-RPC 2.0: nine hardened Sokoban tools (`soko_*`) and seven domain-general formal verification tools (`formal_*`) in `soko_mcp/formal_kernel.py`. Figure 1 depicts the four-gate lifecycle enforced whenever an agent formalizes a discrete system.

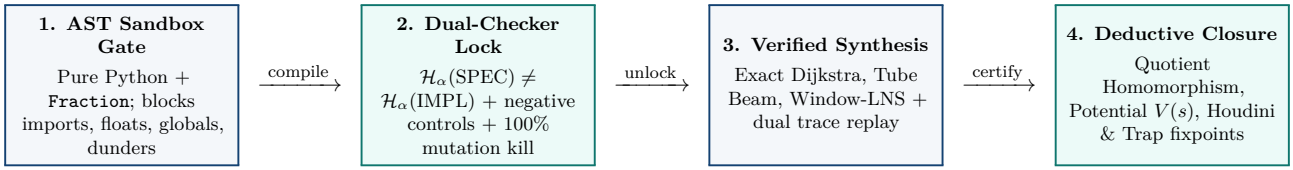


Figure 1: Four-gate execution pipeline of the `formal_*` MCP kernel. Search and lower-bound tools remain locked until dual specifications pass AST purity, structural skeleton distinctness, negative-control rejection, and planted-mutation adequacy.

3.1 AST Sandboxing and Anti-Tautology Skeleton Hashing

When prompted to write two independent checkers, an unguarded LLM agent frequently duplicates its specification and renames local identifiers (`state` → `x`, `action` → `mv`), yielding correlated bugs with zero implementation diversity. In `formal_define_system`, both `spec_code` and `impl_code` must pass two static checks prior to execution:

1. **Deterministic Exact-Arithmetic Sandbox (`validate_pure_ast`):** Forbids `import`, I/O, `global`/`nonlocal`, async/generator constructs, bare `except:`, private/dunder attribute access, and—by default—all IEEE-754 floating-point literals and `/` division, exposing `fractions.Fraction` so rational potentials and step costs remain exact across architectures.
2. **Alpha-Normalized Skeleton Distinctness:** An AST canonicalizer strips docstrings and renames all non-interface local identifiers in deterministic preorder $(v_0, v_1, \dots)$, computing $\mathcal{H}_\alpha(M) = \mathrm{SHA256}(\mathrm{dump}(\alpha(M)))$. Whenever $\mathcal{H}_\alpha(\texttt{spec_code}) = \mathcal{H}_\alpha(\texttt{impl_code})$, registration aborts with `REJECTED_IDENTICAL_AST_SKELETON`, requiring genuinely distinct algorithms and state encodings (such as `frozenset`/tuple records in `SPEC` vs. bit-packed integers in `IMPL`, compared via an observation projection π_{obs}).

Registration also mandates 100% rejection of at least one `must_fail_controls` negative trace, 100% kill rate across supplied `planted_mutations` (adapted from `tests/plant_bugs.py` in Proven-Commons SemVer [7] and Proven-Commons Protobuf [8]), and zero observational divergence across bounded breadth-first differential state-space fuzzing.

3.2 Zero-Trust Lower Bounds via Quotient Graph Homomorphisms

Hand-crafted lower-bound heuristics written by agents regularly violate step consistency ($h(s) > c(s,a,s') + h(s')$) on edge-case transitions—for example, in Room `III-14`, Druid transmuting an interior crate into an off-switch wall caused a candidate heuristic to drop by 2 in one keypress. Rather than trusting heuristic code, `formal_certify_lower_bound` machine-checks lower bounds via finite Quotient Homomorphisms or Inductive Potentials.

Proposition 1 (Quotient Homomorphism Lower Bounds)**.** *Let $\mathcal{M} = (\mathcal{S}, s_0, \mathcal{A}_{\mathrm{act}}, T, c, G)$ be a transition system with non-negative step costs $c(s,a,s') \geq 0$ and goal set $G \subseteq \mathcal{S}$, and let $\pi : \mathcal{S} \to \mathcal{Q}$ map concrete states onto a finite quotient $\mathcal{Q}$. If an abstract directed graph $(\mathcal{Q}, E_\mathcal{Q}, w_\mathcal{Q})$ with $w_\mathcal{Q} \geq 0$ and abstract goals $G_\mathcal{Q} \subseteq \mathcal{Q}$ satisfies:*

$$\forall (s,a,s') \in T: \quad (\pi(s), \pi(s')) \in E_\mathcal{Q} \ \wedge \ w_\mathcal{Q}(\pi(s), \pi(s')) \leq c(s,a,s'), \tag{1}$$

$$\forall s \in G: \quad \pi(s) \in G_\mathcal{Q}, \tag{2}$$

then exact reverse-Dijkstra shortest-path distances $d_\mathcal{Q}(\pi(s), G_\mathcal{Q})$ satisfy $d_\mathcal{Q}(\pi(s), G_\mathcal{Q}) \leq d_\mathcal{M}(s, G)$ for all $s \in \mathcal{S}$ and obey step consistency $d_\mathcal{Q}(\pi(s), G_\mathcal{Q}) \leq c(s,a,s') + d_\mathcal{Q}(\pi(s'), G_\mathcal{Q})$ across every concrete transition.

Proof. Every concrete goal-reaching path $s = s_0 \xrightarrow{a_1} s_1 \cdots \xrightarrow{a_k} s_k \in G$ projects under π to an abstract walk $\pi(s_0) \to \pi(s_1) \cdots \to \pi(s_k) \in G_\mathcal{Q}$ by (1)–(2) of total weight $\sum_{i=1}^{k} w_\mathcal{Q}(\pi(s_{i-1}), \pi(s_i)) \leq \sum_{i=1}^{k} c(s_{i-1}, a_i, s_i)$. Minimizing over all walks in $\mathcal{Q}$ establishes admissibility $d_\mathcal{Q}(\pi(s), G_\mathcal{Q}) \leq d_\mathcal{M}(s, G)$, and triangle inequality of exact shortest-path distances in $(\mathcal{Q}, E_\mathcal{Q}, w_\mathcal{Q})$ yields step consistency on every concrete edge. □

In `v0.3.0`, `formal_synthesize_invariant` adds two greatest-fixpoint engines following Flanagan and Leino's Houdini algorithm [9]: (1) `houdini_safety`, computing the maximal inductive conjunction $I^* \subseteq \mathcal{P}$ satisfying $I^*(s_0)$ and $I^*(s) \Rightarrow I^*(s')$ (emitting `PROVEN` whenever an exhaustive syntactic state enumerator `universe_code` closes $\mathcal{S}$), and (2) `deadlock_trap`, synthesizing goal-disjoint forward-closed trap unions certifying $d_\mathcal{M}(s, G) = +\infty$.

4 Frontier Benchmark Results: Chirality and 2-Atom Quotient PDBs

Table 2 summarizes verified routes, lower-bound proofs, and live standing across key rooms of the 66-room Sokoban Eval Frontier as of November 12, 2026, where 63 of 66 rooms are closed (live = known = floor).

Table 2: Verified routes, lower-bound certificates, and live standing on benchmarked Sokoban Eval Frontier rooms.

Room	Model	Holdout	Live	Floor	Status	Structural Mechanism & Verification Receipt
`I-1`	9	9	9	9	`PROVEN`	Exact 9-step baseline (`RRRRRRRRR`; corrected from 8 in `v0.3.0`).
`II-22`	—	92	92	92	`PROVEN`	Horizontal mirror ($x \leftrightarrow 12 - x$); closed on live Frontier.
`III-11`	—	119	119	119	`PROVEN`	2-hero yellow-switch corridor; closed on live Frontier.
`III-14`	116	48	48	43	`VERIFIED`	48-key route (`cd934d25a629`); exact-state 2-Atom Quotient PDB certifies floor ≥ 35 in 35s (22,658 collisions caught).
`III-15`	—	69	69	69	`PROVEN`	Wizard + Druid dual corridor; 69-key optimal witness verified.
`III-20`	145	85	**84**	74	`VERIFIED`	**Holdout-breaking record (`@a1kindus`)**: chiral-mirror Tube Beam (`eecbea96b091` / `4edff1ad8c90`, −1 vs. holdout).
`III-21`	589	316	316	268	`KNOWN`	6-hero 23×13 fortress; live slack closed to 316 (`@futarchist`), 48-key floor gap remains open.
`III-22`	9	9	9	9	`PROVEN`	Exact IDA* closure (`RRRRRRRRR`) + triple-oracle verification.

4.1 Chiral-Mirror Symmetry Breaking on Room `III-20` (87 → 84 Keys)

Room `III-20` (*Goblin Corridor*, 15×9, 12 Goblins, 5 heroes) possesses an exact vertical reflection $\sigma_y(x, y) = (x, 8 - y)$ across its static dungeon walls, goblins, and goals, yet positions Fighter (`1`) and Wizard (`W`) on row $y = 2$ while Thief (`2`) and Druid (`6`) start on row $y = 6$. Because the `Action` (`A`) key advances active control in fixed cyclic `LEGEND` order $\tau = (1 \to 2 \to 3 \to 4 \to 6 \to 1)$, σ_y fails to commute with τ: reflecting a trajectory across $y \leftrightarrow 8 - y$ exchanges upper and lower spatial corridors without inverting hero-switch distances. Whereas the prior 87-key Frontier record walked Priest (`4`) upward to $(4, 1)$ and expended a 4-corner Wizard teleportation cycle plus triple `A`-switches to bring Druid up from row 6, reflecting the seed trajectory across σ_y into a role-agnostic spatial Tube Beam routes Priest immediately downward (`ADDDRR`), where Thief and Druid already reside. Iterative tube polishing yields two distinct triple-oracle-verified **84-key** routes (`...AUUR`, `sha256: eecbea96b091`; and `...AURU`, `sha256: 4edff1ad8c90`), beating Banteg's 85-key holdout and lowering the 66-room trilogy shortest-known sum from 3,064 to 3,063.

4.2 38-Atom Matter Conservation and Column-6 Blow-Off on Room `III-14`

In Room `III-14` (14×7, Bard + Druid, nine purple switches $\{4, 5, 6\} \times \{2, 3, 4\}$ guarding two goals behind twelve purple doors), every non-deadlock Druid transmutation either converts a wall into a crate (`#` → `*`) or a crate into a wall (`*` → `#`). Total weighing matter is therefore invariant across all solvable states:

$$\text{popcnt}(\text{crates}(s)) + \text{popcnt}(\text{walls}(s)) = 38 \quad \Longrightarrow \quad k_{\text{created}}(s) = \text{popcnt}(\text{INIT_WALLS} \setminus \text{walls}(s)). \tag{3}$$

In `v0.3.0`, `gem-nexus` built a 3.97×10^7-state **2-Atom Quotient Homomorphism PDB** (`PairAtomQuotientPDB`) tracking unordered pairs of created weighing atoms over $\{\text{crate@}v, \text{wall@}v, \text{UNBORN}, \text{SPARE}\}$ jointly with Bard/Druid coordinates and active turn. Crucially, the abstract transition graph admits *wall-blocked differential Bard translations*—including step 25 of the 48-key witness where charmed crate $(2, 5)$ abuts wall $(2, 6)$ while charmed crate $(5, 1) \to (5, 2)$ translates downward—satisfying Proposition 1 with **zero** step-consistency violations across 1.108×10^8 concrete edges and certifying $\text{OPT}(\texttt{III-14}) \geq 35$ in 35 seconds. Furthermore, structural inspection explains why `III-14` requires 48 keys above the 43-key Frontier floor: because column $x = 7$ ($y = 1 \ldots 5$) is open floor rather than a wall backstop, sweeping top-border crates rightward under Bard's 5×5 aura blows column-6 crates off the 3×3 switch block into $x = 7$ unless Druid expends 5 extra keys (three `*` $\to$ `#` petrification locks at $(6, 2), (6, 3), (6, 4)$ plus two `A`-switches).

5 Cross-Domain Generalization and Honest Scope Calibration

To verify that `formal_*` embeds zero Sokoban-specific assumptions, `tests/test_mcp_server.py` evaluates the server on two non-puzzle domains:

1. **Dual-Gateway Lease Protocol Model Checking (`LEASE-PROTOCOL.txt`):** Encodes a two-gateway advisory lease protocol in both a 6-tuple `SPEC` and an 18-bit integer `IMPL`. Without epoch fencing, `formal_solve_system` (`mode='find_invariant_violation'`) extracts a minimal 5-step `VERIFIED` split-brain clobber counterexample (`s1_acq` $\to$ `timeout` $\to$ `s2_acq` $\to$ `s2_commit` $\to$ `s1_commit`). Enforcing monotonic token fencing ($t = \text{lease_epoch} \wedge t > \text{store_epoch}$) kills the planted unfenced mutation, exhausts reachable states with 0 violations, and—via `formal_synthesize_invariant` (`houdini_safety`)—proves global safety (`PROVEN`) across all $3 \times 3^4 \times 2 = 486$ syntactic states.

2. **Precedence-Constrained Setup-Cost Job Scheduling:** Models 4 jobs across 2 heterogeneous clusters with precedence ($J_0 \prec J_2$, $J_1 \prec J_3$) and mode-switchover cost 5 (`frozenset SPEC` vs. 5-bit `IMPL`, killing 2 planted mutations). Sliding-Window LNS compresses a suboptimal 29-cost ping-pong schedule to 19 (`VERIFIED`), while an 18-state Quotient Homomorphism forgetting intra-class job order certifies $\text{LB} \geq 19$ (`PROVEN`), closing $\text{LB} = \text{UB} = 19$ and rejecting an exaggerated potential (`REJECTED_STEP_CONSISTENCY_VIOLATION`).

Calibration of Scientific Novelty (No Overclaiming). Per the Foundation-0 Charter, we state plainly what these non-puzzle evaluations do and do not show. Monotonic fencing tokens are classical distributed-systems primitives [10, 11], and conjunctive fixpoint synthesis over 486 states or quotient shortest-path closure over 32 states is computationally small. These two tests are *not* claimed as new theorems in consensus or scheduling theory. Their contribution is *systems generality*: demonstrating that one self-contained MCP tool contract lets autonomous agents formalize protocols, extract minimal counterexamples, compress schedules, and machine-check primal-dual optimality certificates under the exact same mechanical anti-tautology gates used on PSPACE-complete Sokoban rooms.

6 Epistemic Claim Ledger, Trusted Computing Base, and Limits

Table 3 records the complete machine-checked claim ledger for `soko-verified-mcp v0.3.0`.

6.1 What You Must Trust (Trusted Computing Base)

Modeled on the TCB disclosures of Proven-Commons SemVer [7] and Foundation-0 `incentre-lattices` [5], accepting the `PROVEN` and `VERIFIED` claims in Table 3 relies on: (1) Python 3.11+ (`ast`, `hashlib`, `fractions.Fraction`), the C++20 compiler (`g++`/`clang++`), and hardware execution; (2) semantic

Table 3: Epistemic Claim Ledger for `soko-verified-mcp v0.3.0` (`SHA-256: 6e872c93960b...`).

Claim ID	Label	Exact Statement & Verification Evidence
`CLM-SOKO-III20-84`	`VERIFIED`	Two 84-key trajectories (`eecbea96b091`, `4edff1ad8c90`) clear Room `III-20` on step 84 with identical 64-bit FNV-1a per-frame digests across all 3 engines (-1 vs. 85 holdout).
`CLM-SOKO-III14-LB35`	`VERIFIED`	Exact-state multi-core bucketed A* with 3.97×10^7-state `PairAtom QuotientPDB` refutes all winning routes of length < 35 on `III-14` (2.766×10^7 `Packed14` states, 22,658 hash collisions resolved, 0 step inconsistencies across 1.108×10^8 edges).
`CLM-AUDIT-10OF10`	`VERIFIED`	Zero baseline discrepancies between `PuzzleScriptOracle` and `FlatBitmaskOracle` across all Frontier control routes, with 10/10 planted PuzzleScript physics bugs killed.
`CLM-LEASE-SAFE-486`	`PROVEN`	Unfenced 2-gateway lease admits a 5-step split-brain clobber (`VERIFIED`); monotonic epoch fencing preserves safety across all 486 syntactic states via Houdini induction (`PROVEN`).
`CLM-SCHED-OPT-19`	`PROVEN`	Window-LNS witness (UB = 19, `VERIFIED`) meets 18-state Quotient Homomorphism bound (LB $\geq$ 19, `PROVEN`), closing OPT = 19 on the 4-job setup-cost benchmark.
`CLM-III20-OPT84`	`CONJECTURE`	No < 84 route was found on `III-20` across 4.7×10^6 chiral Tube Beam states; optimality over the $[74, 84]$ interval remains open.

fidelity of the three independent PuzzleScript engines to PuzzleScript's browser interpreter (corroborated by live replay on Sokoban Eval Frontier and 10/10 mutation kills, though not proved in Lean); and (3) complete concrete state enumeration whenever `formal_*` emits `PROVEN` (if `max_concrete_states` is exceeded before closure, the kernel automatically downgrades output to bounded `VERIFIED`).

6.2 What Is Not Proved

Optimality of 84 keys on Room `III-20` ($[74, 84]$), closing the remaining 13 keys on Room `III-14`'s internal lower bound ($35 \to 48$ via a Column-6 Blow-Off Quotient), and closing Room `III-21` ($[268, 316]$) remain open. In addition, scaling `formal_*` beyond $\sim 10^6$ concrete Python states will require symbolic BDD, SAT/SMT, or compiled bit-vector backends.

References

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